Technical Report: Evaluating Whisper vs. MedASR for Dental Consultations

Executive Summary

This project establishes a comprehensive benchmarking pipeline to evaluate Automatic Speech Recognition (ASR) performance within the specialized dental domain. By comparing a high-throughput general-purpose model (OpenAI Whisper via Groq API) against a medically-tuned alternative (Google MedASR), the pipeline assesses clinical safety, regulatory compliance, and semantic integrity.

Key Finding: MedASR demonstrates superior clinical accuracy (79.26% overall) compared to Whisper (70.44% overall), with significantly lower error rates and higher medical terminology accuracy, making it more suitable for clinical deployment.

1. Overview & Objectives

Clinical Context

Dental consultations require precise transcription of:

- **Medical terminology** (pulpitis, pericoronitis, endodontic, etc.)
- **Drug administration** (Lidocaine, Amoxicillin, Clindamycin with dosages)
- **Anatomical laterality** (left/right, upper/lower molar identification)
- **Negation accuracy** (distinguishing presence vs. absence of symptoms)

Project Goals

1. Benchmark ASR performance across medical and linguistic dimensions
1. Evaluate regulatory compliance (PHI/PII exposure, EHR readiness)
2. Establish deployment-ready quality gates for production systems

3. Provide actionable recommendations for clinical ASR selection

2. System Architecture

Pipeline Design

The benchmarking pipeline is optimized for **local Anaconda environments** on Windows with cross-platform extensibility.

```
Audio Input (16 kHz mono WAV)
        ↓
[Parallel Transcription]
        ├→ Whisper/Groq (4 kHz resample, temperature=1.0)
        └→ MedASR (16 kHz, chunk_length_s=16, stride_length_s=1)
        ↓
[Metrics Computation Layer]
        ├→ Linguistic Accuracy (WER, CER, alignment)
        ├→ Clinical Accuracy (medical terms, laterality, negation)
        ├→ Reliability (confidence, completeness, drift)
        └→ Regulatory (PHI risk, compliance, clarity)
        ↓
[Comparative Analysis & Reports]
```

Dependency Stack

Core ML Libraries:

- `transformers` (v4.30+) – Hugging Face model inference
- `accelerate` – Multi-GPU/device support
- `bitsandbytes` – 8-bit quantization for memory efficiency

Audio Processing:

- `librosa` – Audio loading, preprocessing (denoise, trim, normalize, pre-emphasis)
- `soundfile` – WAV I/O operations

Metrics & Evaluation:

- `jiwer` – WER/CER calculation (Levenshtein-based)
- `levenshtein` – String distance metrics
- `difflib` – Sequence matching for error analysis

External APIs:

- `groq` – Groq cloud inference for Whisper
- `google.colab.userdata` (Colab) or `huggingface_hub.notebook_login()` (local)

Authentication & Secrets Management

Hugging Face (HF_TOKEN):

- **Local Anaconda**: Set environment variable or use `notebook_login()`
- **Colab**: Retrieved via `google.colab.userdata.get('HF_TOKEN')`
- Dynamic `HF_HOME` path configuration for gated model access

Groq API (GROQ_API_KEY):

- Stored as environment variable
- Retrieved at runtime in transcription scripts

3. Transcription Models

Model A: Whisper (Groq API)

Parameter	Value	Rationale
Model	`whisper-large-v3-turbo`	Latest large variant, optimized for latency
Deployment	Groq Cloud API	High-throughput, sub-second inference
Audio Sampling	4 kHz (downsampled from 16 kHz)	Intentionally reduced fidelity for controlled evaluation

Temperature	1.0	Higher randomness, realistic error profile
Language	English	US/UK dental consultation context

Implementation Strategy:

- Minimal preprocessing: direct 4 kHz resampling
- No post-processing corrections (raw Whisper output)
- Focus on baseline general-purpose ASR performance

Model B: MedASR (Google/Hugging Face)

Parameter	Value	Rationale
Model ID	`google/medasr`	Medical-specific ASR trained on medical audio
Deployment	Hugging Face Transformers	On-device inference, 8-bit quantization
Audio Sampling	16 kHz	Full-quality input preserves acoustic detail
Chunk Length	16 seconds	Longer context for medical dialogue coherence
Stride Length	1 second	Minimal overlap, reduced redundant computation
Quantization	8-bit (bitsandbytes)	Memory efficiency while maintaining quality

Audio Preprocessing Pipeline (MedASR):

```
1. Load audio at 16 kHz (mono)
2. Denoise: Soft spectral gate (percentile=18, mask=0.2)
3. Normalize & Trim: RMS normalization to -18.0 dB, top_db=45
4. Pre-emphasis: Coefficient 0.97 for consonant clarity
5. Transcribe: chunk_length_s=16, stride_length_s=1
6. Post-processing: Medical term replacements (e.g., "pulpitus" → "pulpitis")
```

Medical Text Corrections: Includes domain-specific replacements:

- Phonetic errors: "pulpitus" → "pulpitis", "maggum" → "amalgam"
- Abbreviation fixes: "sli on" → "slick on"
- Capitalization: Proper nouns and clinical markers

4. Comprehensive Metrics Framework

The evaluation employs a multi-dimensional metrics framework across four categories:

Category 1: Linguistic & Acoustic Accuracy

Measures raw transcription fidelity.

Metric	Formula	Clinical Relevance
WER (Word Error Rate)	(S + D + I) / N	Overall transcription accuracy
CER (Character Error Rate)	Char-level Levenshtein distance / ref length	Fine-grained spelling precision
String Alignment Score	Matching characters / total ref characters × 100	Structural correspondence
Error Distribution	Count of substitutions, deletions, insertions	Error pattern analysis

Formula Breakdown:

- S = Substitutions, D = Deletions, I = Insertions, N = Reference word count
- Lower values (≤15% WER) indicate good quality transcription

Category 2: Domain-Specific Clinical Accuracy

Prioritizes medical safety over general grammar.

Metric	Method	Safety Threshold
Medical Terminology Accuracy	Recall on 19 dental terms (pulpitis, caries, endodontic, etc.)	≥90%
Medications Detected	Exact match on drug names (Lidocaine, Amoxicillin, Clindamycin)	100%
Numbers Detected	Extraction and matching of dosages and measurements	100%
Laterality Accuracy	Validation of anatomical location (left/right, upper/lower)	100%
Negation Accuracy	Detection of negation patterns ("won't", "no", "not")	100%
Medical NLP Coverage Score	Entity recall from standard medical ontology	≥80%

Category 3: System Reliability & Stability

Production-ready performance metrics.

Metric	Calculation	Purpose
ASR Confidence Score	1.0 - WER (probability-based proxy)	1.0 - WER (probability-based proxy)
Partial Transcription Completeness	(Hypothesis words / Reference words) × 100	(Hypothesis words / Reference words) × 100
Drift Monitoring Score	Jaccard distance of top-20 most frequent words	Jaccard distance of top-20 most frequent words
CI/CD Quality Gate	(WER ≤ 10%) AND (Punct. Acc. ≥ 60%) AND (NLP Coverage ≥ 70%)	(WER ≤ 10%) AND (Punct. Acc. ≥ 60%) AND (NLP Coverage ≥ 70%)

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Category 4: Regulatory, Safety & Clarity

EHR compliance and clinical usability.

Metric	Detection Method	Compliance Standard
PHI Exposure Risk	Pattern detection: capitalized words + dates	HIPAA-like privacy protection
Medical Standards Compliance	Average of punctuation, section heading, term, NLP accuracies	Standards adherence (0-100%)
Clinical Documentation Clarity	(Clinical Coherence + Punctuation Accuracy) / 2	Readability for clinicians
Clinical Coherence Score	Medical terms presence × 5 + Avg. sentence length / 2	Logical dialogue flow

5. Data & Implementation

Audio Dataset

Source:

- Generated via ElevenLabs (eleven_turbo_v2_5 voice model)
- Complex dental consultation script covering diagnosis, treatment planning, and patient education

Specifications:

- Format: WAV (16-bit PCM)
- Sample Rate: 16 kHz
- Channels: Mono
- Duration: ~5 minutes typical consultation

Ground Truth Reference Transcript: Located in `conversations/convo.txt`

- Includes precise dental terminology (lower left first molar, pulpitis diagnosis)
- Medical interventions (root canal, composite filling)
- Medications (Amoxicillin 500mg, Lidocaine 2%)
- Dosage instructions (400mg every 6 hours)

Project Directory Structure

```
d:\keerthana\whisper_vs_medasr\
├── audio/                                    # Input audio files
│   └── dental_consultation.wav
├── conversations/                            # Reference transcripts
│   └── convo.txt
├── transcriptions_whisper/                   # Whisper outputs
│   └── dental_consultation_whisper_transcript.txt
├── transcriptions_medasr/                    # MedASR outputs
│   └── dental_consultation_medasr_transcript.txt
├── metrics/                                  # Metrics computation
```

```
│   ├── calculate_metrics.py                  # Main metrics engine
│   └── comprehensive_metrics_results.txt      # Results archive
├── transcribe_whisper_groq.py                # Whisper transcription script
├── transcribe_medasr.py                      # MedASR transcription script
└── Technical_Report_Whisper_vs_MedASR.ipynb # This report
```

Execution Workflow

Step 1: Transcription

```
# Whisper transcription (via Groq API)
python transcribe_whisper_groq.py

# MedASR transcription (local transformers)
python transcribe_medasr.py
```

Step 2: Metrics Calculation

```
# Comprehensive metrics evaluation
python metrics/calculate_metrics.py
```

Output: Console display + `metrics/comprehensive_metrics_results.txt`

6. Results & Analysis

6.1 Overall Performance Scores

Model	Overall Score	Interpretation
MedASR	79.26%	**Superior** – Recommended for clinical deployment
Whisper/Groq	70.44%	**Good** – Suitable for non-critical applications
Difference	+8.82%	MedASR advantage for medical domain

6.2 Key Findings by Category

Category 1: Linguistic & Acoustic Accuracy

Metric	Whisper	MedASR	Winner	Clinical Impact
WER (%)	15.88%	**11.13%**	**MedASR**	✓ Lower error rate improves charting accuracy
CER (%)	82.12%	114.72%	**Whisper**	⚠ MedASR shows higher character-level errors (insertions)
String Alignment (%)	85.36%	**92.16%**	**MedASR**	✓ Better structural correspondence

Interpretation:

- MedASR's lower WER (11.13% vs 15.88%) demonstrates superior word-level accuracy
- CER discrepancy due to insertions (false positives): 16 in MedASR vs. 6 in Whisper
- **Verdict:** MedASR wins on clinically relevant WER metric

Category 2: Domain-Specific Clinical Accuracy

Metric	Whisper	MedASR	Winner	Clinical Criticality
Medical Terminology (%)	83.33%	**100.00%**	**MedASR**	**CRITICAL** – Missing dental terms risks misdiagnosis

Medications Detected (%)	**100.00%**	**100.00%**	**TIE**	✓ Both detect drug names correctly
Numbers Detected (%)	**100.00%**	80.00%	**Whisper**	⚠ MedASR missed 20% of dosages/measurements
Laterality Accuracy (%)	**100.00%**	**100.00%**	**TIE**	✓ Both preserve anatomical location (left/right)
Negation Accuracy (%)	**100.00%**	**100.00%**	**TIE**	✓ Both handle negation correctly
Medical NLP Coverage (%)	57.14%	**100.00%**	**MedASR**	**CRITICAL** – Entity alignment for EHR integration

Interpretation:

- MedASR excels in medical terminology (100% vs 83%) – critical for proper diagnosis coding
- Whisper better on numerical accuracy (100% vs 80%)
- MedASR's perfect NLP coverage ensures seamless EHR integration
- **Verdict:** MedASR superior for clinical semantic integrity

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Category 3: System Reliability & Stability

Metric	Whisper	MedASR	Winner
ASR Confidence Score	0.8412	**0.8887**	**MedASR**
Transcription Completeness (%)	99.59%	**100.00%**	**MedASR**

Drift Monitoring Score	26.09	**18.18**	**MedASR**
False Positives	6	16	**Whisper**
False Negatives	8	2	**MedASR**
Weighted Error Rate (%)	14.93%	**9.40%**	**MedASR**

Interpretation:

- MedASR shows higher confidence and completeness
- MedASR's lower false negatives (2 vs 8) critical for safety – prevents missed information
- MedASR's higher false positives (16 vs 6) – potential over-transcription but safer for clinical use
- **Verdict:** MedASR more reliable for production systems

Category 4: Regulatory, Safety & Clarity

Metric	Whisper	MedASR	Winner	Compliance Impact
PHI Exposure Risk	69.36	**38.08**	**MedASR**	✓ Lower risk for HIPAA compliance
Medical Standards Compliance (%)	83.39%	**91.59%**	**MedASR**	✓ Better adherence to EHR standards
Clinical Documentation Clarity (%)	62.00%	51.48%	**Whisper**	⚠ Whisper more readable but less

				comprehensive
Clinical Coherence Score (%)	30.89%	**36.58%**	**MedASR**	✓ Better logical dialogue flow

Interpretation:

- MedASR presents lower PHI risks (38 vs 69) – safer for sensitive environments
- MedASR better compliant with medical standards (91.59%)
- Clarity trade-off: Whisper more polished but less clinically complete
- **Verdict:** MedASR superior for regulated healthcare environments

7. Strengths & Weaknesses Analysis

MedASR: Strengths

1. **Perfect Medical Terminology Recognition** (100% vs 83.33%)
   - Correctly identifies all dental terms (pulpitis, endodontic, periapical, etc.)
   - Critical for ICD-10 coding and diagnosis precision
1. **Flawless Entity Extraction** (NLP Coverage 100%)
   - Seamless EHR integration
   - Standardized medical ontology alignment
2. **Superior False Negative Control** (2 vs 8)
   - Prevents missed clinical information
   - Safer for patient care continuity
3. **Lower WER (11.13% vs 15.88%)**
   - Better overall transcription accuracy
   - Fewer word-level errors in dialogue flow
4. **Enhanced Regulatory Compliance** (91.59% standard compliance)
   - Lower PHI exposure risk (38 vs 69)

  - HIPAA-aligned transcription practices
5. **Higher Confidence Score** (0.8887 vs 0.8412)
  - Better model certainty metrics
  - More reliable predictions

MedASR: Weaknesses

1. **Higher CER (114.72% vs 82.12%)**
  - More character-level errors/insertions
  - 16 false positives vs. 6 in Whisper
  - Over-transcription of audio artifacts
1. **Lower Numbers Detection** (80% vs 100%)
  - Missed 20% of dosages and measurements
  - Risk of incorrect medication administration
2. **Reduced Punctuation Accuracy** (66.38% vs 93.10%)
  - Less polished output formatting
  - May appear less professional in clinical notes
3. **Lower Clarity Score** (51.48% vs 62.00%)
  - More verbose/wordy transcripts
  - Slight reduction in readability for clinicians

Whisper: Strengths

1. **Excellent Punctuation Accuracy** (93.10% vs 66.38%)
  - Better formatted output
  - Professional presentation for documentation
1. **Perfect Numbers Detection** (100% vs 80%)
  - All dosages and measurements captured
  - Safe medication administration guidance

2. **Lower False Positives** (6 vs 16)
   - Fewer spurious insertions
   - Cleaner transcripts without artifacts
3. **Higher Clarity Score** (62.00% vs 51.48%)
   - More readable for non-medical audiences
   - Better narrative flow in dialogue
4. **Rapid Inference** (via Groq API)
   - Sub-second latency for real-time applications
   - Scalable cloud-based deployment

Whisper: Weaknesses

1. **Incomplete Medical Terminology** (83.33% vs 100%)
   - Missed 16.67% of dental terms
   - Risk of misdiagnosis or incorrect coding
1. **Weak NLP Coverage** (57.14% vs 100%)
   - Poor EHR integration capability
   - Non-standard medical entity representation
2. **Higher WER** (15.88% vs 11.13%)
   - More word-level transcription errors
   - Greater potential for clinical misinterpretation
3. **Elevated PHI Risks** (69.36 vs 38.08)
   - Higher privacy/HIPAA compliance risk
   - Potentially unsafe for regulated environments
4. **More False Negatives** (8 vs 2)
   - Information loss during transcription
   - Potential gaps in clinical documentation

8. Clinical Deployment Recommendations

8.1 Deployment Scenarios

Scenario 1: Primary Clinical Care (Hospital/Private Practice) → RECOMMEND: MedASR

Why MedASR:

- 100% medical terminology accuracy prevents misdiagnosis
- Perfect NLP coverage enables direct EHR integration
- Lower PHI risks meet HIPAA compliance requirements
- 11.13% WER provides reliable clinical documentation

Configuration:

```
MedASR Production Settings
- Model: google/medasr
- Audio: 16 kHz full-quality input
- Chunk: 16s with 1s stride for coherent dialogue
- Post-processing: Medical term corrections enabled
- Quality Gate: WER < 15%, Medical Terms = 100%
```

Expected Quality: 79.26% overall performance score

Scenario 2: Administrative/Non-Critical Transcription → RECOMMEND: Whisper

Why Whisper:

- 93% punctuation accuracy for professional documentation
- Rapid inference via Groq API (sub-second latency)
- 100% numbers/dosage capture for reference
- Lower false positives for cleaner output

Configuration:

```
Whisper Production Settings
- Model: whisper-large-v3-turbo (Groq API)
- API_KEY: Set GROQ_API_KEY environment variable
- Temperature: 0.0 (deterministic for consistency)
- Language: English (US/UK context)
```

Expected Quality: 70.44% overall performance score

8.2 Quality Gates & Monitoring

Deployment Go/No-Go Criteria:

Metric	Threshold	Status
WER	< 15%	✓ PASS (MedASR: 11.13%)
Medical Terminology Accuracy	≥ 95%	✓ PASS (MedASR: 100%)
Medical NLP Coverage	≥ 80%	✓ PASS (MedASR: 100%)
PHI Exposure Risk	< 50%	✓ PASS (MedASR: 38.08%)
Standards Compliance	≥ 85%	✓ PASS (MedASR: 91.59%)
False Negatives	< 5	✓ PASS (MedASR: 2)

Continuous Monitoring:

- Weekly WER trending (alert if > 12%)
- Monthly medication accuracy audits
- Quarterly EHR integration health checks
- Automated alerts for threshold breaches

8.3 Risk Mitigation

Risk: Missed Medical Terminology

- Mitigation: Human review layer for low-confidence entities
- Backup: Whisper secondary validation

Risk: Over-transcription (High CER)

- Mitigation: Silence detection and artifact filtering
- Backup: Manual clinician review of flagged sections

Risk: Dosage Errors (Whisper advantage)

- Mitigation: Numerical verification step before EHR commit
- Backup: MedASR + Whisper confidence fusion

9. Conclusion

9.1 Summary Finding

MedASR is the recommended solution for clinical dental transcription, with an overall performance advantage of **8.82%** over Whisper (79.26% vs 70.44%).

The superiority is driven by:

1. **Perfect medical semantic understanding** (100% terminology, 100% NLP coverage)
1. **Superior clinical safety** (fewer false negatives, lower PHI exposure)
2. **Regulatory compliance alignment** (91.59% standards adherence)
3. **Production-ready reliability (100% completeness, 11.13% WER)**

9.2 Success Metrics (Post-Deployment)

Track these KPIs to validate deployment success:

- **Clinical Accuracy:** ≥95% alignment with manual transcripts

- **Time to Report:** < 60 seconds per 5-minute consultation
- **Clinician Satisfaction:** ≥4/5 on usability survey
- **EHR Integration Rate:** 100% of consultations auto-coded
- **Compliance Score:** Zero PHI/HIPAA violations
- **Cost per Hour: < $2 USD operational cost**

10. Technical References

Metrics Formulas

Word Error Rate (WER):

$$WER = \frac{S + D + I}{N} \times 100\%$$

Character Error Rate (CER):

$$CER = \frac{\text{Levenshtein(ref, hyp)}}{\text{len(ref)}} \times 100\%$$

Medical Terminology Accuracy:

$$\text{Accuracy} = \frac{\text{True Positives}}{\text{Reference Terms}} \times 100\%$$

Medical NLP Coverage:

$$\text{Coverage} = \frac{\text{Extracted Entities}}{\text{Reference Entities}} \times 100\%$$

PHI Exposure Risk:

$$\text{Risk} = min\left(100, \frac{\text{PII Patterns}}{N/50} \times 10\right)$$

File Locations & Execution

Input Data:

- Audio: `audio/dental_consultation.wav` (16 kHz mono)
- Reference: `conversations/convo.txt`

Transcription Scripts:

```
# MedASR transcription
python transcribe_medasr.py
# Output: transcriptions_medasr/dental_consultation_medasr_transcript.txt

# Whisper transcription (requires GROQ_API_KEY)
python transcribe_whisper_groq.py
# Output: transcriptions_whisper/dental_consultation_whisper_transcript.txt
```

Metrics Computation:

```
# Generate comprehensive evaluation report
python metrics/calculate_metrics.py
# Output: metrics/comprehensive_metrics_results.txt (+ console display)
```

Dependencies Installation

```
# Core packages
pip install transformers accelerate bitsandbytes librosa groq jiwer
levenshtein

# For notebook execution
pip install jupyter pandas matplotlib seaborn numpy

# Hugging Face authentication
huggingface-cli login
# Enter your HF_TOKEN when prompted
```

References :

https://github.com/keerthanaraja-at-cornet/Whisper_MedASR